Experiment 1a)

Program:-

```
#include <stdio.h>
#include <unistd.h>
#include <fcntl.h>

int main()
{
   int fd;
   char buffer[50];

   /* open() – create and open a file */
   fd = open("sample.txt", O_CREAT | O_RDWR, 0644);

   if (fd < 0)
   {
      printf("File opening failed\n");
      return 1;
   }

   /* write() – write data into file */
   write(fd, "Hello OS Lab", 12);

   /* move file pointer to beginning */
   lseek(fd, 0, SEEK_SET);

   /* read() – read data from file */
   read(fd, buffer, 12);

   /* display data */
   printf("Data read from file: %s\n", buffer);

   /* close() – close the file */
   close(fd);

   return 0;
}
```

Experiment 1b)

Program:-

```
#include <stdio.h>
#include <dirent.h>

int main()
{
    DIR *d;
    struct dirent *entry;

    /* open current directory */
    d = opendir(".");

    if (d == NULL)
    {
        printf("Directory cannot be opened\n");
        return 1;
    }
    /* read directory entries */
    printf("Directory contents:\n");
    while ((entry = readdir(d)) != NULL)
    {
        printf("%s\n", entry->d_name);
    }
    /* close directory */
    closedir(d);
    return 0;
}
```

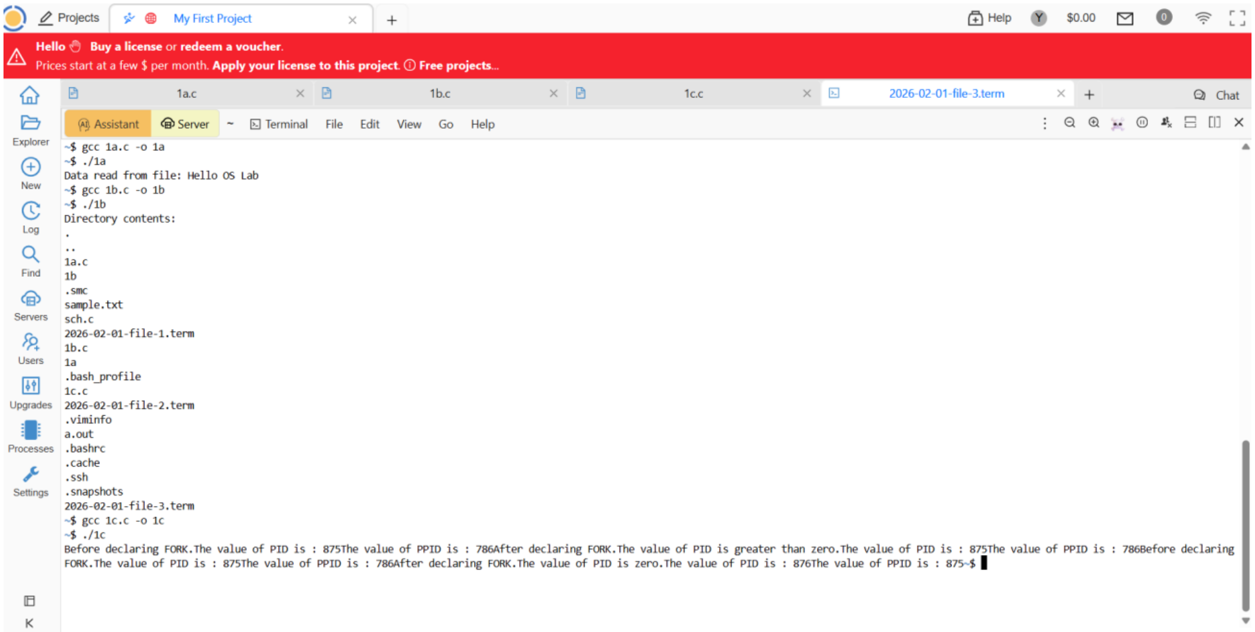
Experiment 1c)

Program:-

```c
#include<stdio.h>
#include<unistd.h>
#include<sys/types.h>
void main()
{
printf("Before declaring FORK.");
printf("The value of PID is : ");
printf("%d",getpid());
printf("The value of PPID is : ");
printf("%d",getppid());

pid_t pid=fork();//Used to create child process
if(pid==0)//Child Process
{
printf("After declaring FORK.");
printf("The value of PID is zero.");
printf("The value of PID is : ");
printf("%d",getpid());
printf("The value of PPID is : ");
printf("%d",getppid());
}
else if(pid>0)//Parent Process
{
printf("After declaring FORK.");
printf("The value of PID is greater than zero.");
printf("The value of PID is : ");
printf("%d",getpid());
printf("The value of PPID is : ");
printf("%d",getppid());
}
else//Error
{
printf("After declaring FORK.");
printf("The value of PID is less than zero.");
printf("The value of PID is : ");
printf("%d",getpid());
printf("The value of PPID is : ");
printf("%d",getppid());
}
}
```

OUTPUT:-


Projects
My First Project
Help
$0.00
Hello Buy a license or redeem a voucher.
Prices start at a few $ per month. Apply your license to this project. Free projects...
1a.c
1b.c
1c.c
2026-02-01-file-3.term
Chat
Assistant
Server
Terminal
File
Edit
View
Go
Help
Explorer
New
Log
Find
Servers
Users
Upgrades
Processes
Settings
~$ gcc 1a.c -o 1a
~$ ./1a
Data read from file: Hello OS Lab
~$ gcc 1b.c -o 1b
~$ ./1b
Directory contents:
.
..
1a.c
1b
.smc
sample.txt
sch.c
2026-02-01-file-1.term
1b.c
1a
.bash_profile
1c.c
2026-02-01-file-2.term
.viminfo
a.out
.bashrc
.cache
.ssh
.snapshots
2026-02-01-file-3.term
~$ gcc 1c.c -o 1c
~$ ./1c
Before declaring FORK.The value of PID is : 875The value of PPID is : 786After declaring FORK.The value of PID is greater than zero.The value of PID is : 875The value of PPID is : 786Before declaring FORK.The value of PID is : 875The value of PPID is : 786After declaring FORK.The value of PID is zero.The value of PID is : 876The value of PPID is : 875~$